IEEE Control Systems Letters paper presented at
2023 American Control Conference (ACC)
San Diego, CA, USA. May 31 - June 2, 2023

Command governor-based adaptive control for constrained linear systems in presence of unmodelled dynamics[†]

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Abstract— **This paper deals with the control of uncertain constrained linear systems. The plant's dominant dynamics are controlled via a simple linear control law, while uncertainties are handled by an adaptive strategy that appropriately modifies the law by monitoring the system state in accordance with some desired dynamics. The closed-loop system is augmented with a robust command governor scheme to enforce state and input constraints by modifying the reference point-wisely. Thanks to the adaptive law performance guarantees, large uncertainties can be handled and the computational complexity of the governor's optimization problem is limited compared to the standard strategy. Stability and feasibility are discussed. Numerical simulations illustrate the methodology applied to a geostationary satellite. Discussions and future perspectives conclude the paper.**

I. Introduction

Complex engineering applications may require dealing with dynamics that cannot be easily modeled and included in a control framework. In addition, an uncertain environment might expose the system to disturbances that are unforeseeable during the control design phase. These scenarios challenge robust controllers, notably conceived to handle precise ranges of perturbations, and might compel them to unnecessary sacrifices on performance to ensure safety and reliability. On the other hand, adaptive laws can ensure excellent tracking performance without prior assumptions about the unknown system dynamics by virtue of their flexibility. However, the typical strong adaptive control action can degrade performance in case the system constraints are not respected. Among the most popular methods to enforce constraints, we find Model Predictive Control (MPC) [1], [2], Reference Governor (RG), and its variation Command Governor (CG) [3], [4]. The three approaches rely on solving an online optimization problem to compute an optimal signal according to the predicted system dynamics. While MPC schemes act in the place of a controller and look for the optimal control input sequence, RGs, and CGs are added between the reference signal and the closed-loop system and filter the desired reference to generate a virtual one whenever constraints violation is at risk. Hence, they do not operate to resolve the system stability and do not interfere with the closed-loop system dynamics. This feature makes RGs and CGs perfectly suit all applications for which an established controller already exists, but no strategy to handle constraints is available. In addition, thanks to the limited computational effort required compared to other optimization-based control devices, such as MPC, reference and command governors can result more appropriate in the applications with fast-dynamics and low computational capability, *e.g.*, aerospace, automotive, and precision mechatronics.

In this work, a solution to control linear uncertain constrained systems is proposed. The objective is twofold: track a reference signal in presence of matched uncertainties, and enforce the system state and input constraints. Under the assumption that the linear dominant plant dynamics are available, a Model Reference Adaptive Control (MRAC) strategy is employed to handle the unknown system dynamics, [5], [6]. By mimicking a desired reference trajectory, MRAC ensures excellent closed-loop performance in presence of plant parameter variations and/or unmodelled system dynamics. Despite the wide range of complex systems that would benefit from the combination of an adaptive law with a constraints-preventing scheme, the related literature is rather limited and essentially renounces to exploit the full controller adaptability to guarantee constraints enforcement, see [7] and [8]. In this regard, the proposed solution enhances an established MRAC-based control law, proven to guarantee stability, with a command governor scheme. CG does not act directly on the closed-loop dynamics; instead, it evaluates the desired reference signal and predicts the closed-loop system trajectories on a predefined horizon. If constraints are not respected, then CG computes the closest signal to the desired reference that allows constraints enforcement and uses it as a virtual reference in the closed-loop system. The optimization-based process is then repeated at each time step. Although the growing interest in optimization-based methods to enforce constraints, combining adaptive controllers with reference and command governors remains a novel approach, and limited literature on the topic exists. Since a reliable model is crucial for the closed-loop system prediction, several approaches developed robust versions of the standard governors to handle the system uncertainties, [9], [10], [11]. These solutions employ the available knowledge about the uncertainty, notably its magnitude bounds, to predict all possible evolution of the closed-loop state and enforce constraints accordingly. However, this could result in being overly conservative and may not allow for handling unmodelled uncertainties. In this regard, this work proposes a relatively simple control architecture that exploits the

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[†]This work was supported by CNES and ONERA

adaptive control law properties to precisely track a desired reference and to build a robust command governor with limited conservatism. Specifically, we exploit the MRAC bound on the error between the desired trajectory and the actual system evolution, see [12] and [13], to tighten the constraint set and ensure constraints enforcement despite the unmodelled uncertainty.

The paper is organized as follows: Section II formally states the problem and clarifies the controller objectives. Section III describes the proposed control framework. In Section IV, the proposed control architecture is applied to a consequent case study. Finally, in Section V conclusions and future perspectives are provided.

II. PROBLEM STATEMENT

Consider a class of constrained uncertain linear systems represented by

$$\dot{x}(t) = Ax(t) + B(u(t) + d(x(t), t)), \quad x(0) = x_0, \quad t \geq 0, \tag{1}$$

where $x(t) \in \mathbb{R}^n$ is the state vector and is assumed available for feedback, $x(0) \in \mathbb{R}^n$ is the initial state vector, $u(t) \in \mathbb{R}^m$ is the control input, and $d(x(t), t) \in \mathbb{R}^m$ is the matched uncertainty that can depend on both states and time. The matrices $A \in \mathbb{R}^{n\times n}$ and $B \in \mathbb{R}^{n\times m}$ describe the known linear time-invariant dominant dynamics of the plant, and the couple (A, B) is assumed controllable. States and input are subject to constraints, such that

$$x(t) \in \mathcal{X}, \quad u(t) \in \mathcal{U}, \quad \forall t \geq 0, \tag{2}$$

where $\mathcal{X} \subset \mathbb{R}^n$ and $\mathcal{U} \subset \mathbb{R}^m$ are convex and compact sets.

Assumption 1: The uncertainty $d(x(t), t)$ is assumed to be parameterized as

$$d(x(t), t) = W^\top(t)\sigma(x(t)), \tag{3}$$

where $W(t) \in \mathbb{R}^{s\times m}$ denotes a bounded unknown weight matrix (*i.e.*, $||W(t)||_2 \leq w, t \geq 0$) with a bounded time rate of change (*i.e.*, $||\dot{W}(t)||_2 \leq \dot{w}, t \geq 0$), with $w, \dot{w}$ real positive constants and $\sigma(x(t)) : \mathbb{R}^n \to \mathbb{R}^s$ denotes a known basis function of the form $\sigma(x(t)) = [\sigma_1(x(t)), \sigma_2(x(t)), ..., \sigma_s(x(t))]$. Note that, by letting one element of the basis function be a constant (*i.e.*, $\sigma_1(x(t)) = 1$), this parametrization can capture exogenous time-dependent uncertainties.

Problem: Design a control that guarantees the tracking of a reference $r(t) \in \mathbb{R}^r$ in spite of plant uncertain dynamics while respecting state/input constraints.

III. PROPOSED SOLUTION

The proposed control framework is illustrated in Fig. 1. A stabilizing nominal controller for the LTI known plant dynamics of (1) is assumed available. An MRAC algorithm compensates for the system uncertainties, and a CG scheme enforces the closed-loop system input and state constraints. The resulting control signal $u(t)$ is the sum of the nominal and of the adaptive contributions, $u_N(t) \in \mathbb{R}^m$ and $u_A(t) \in \mathbb{R}^m$, respectively:

$$u(t) = u_N(t) + u_A(t). \tag{4}$$

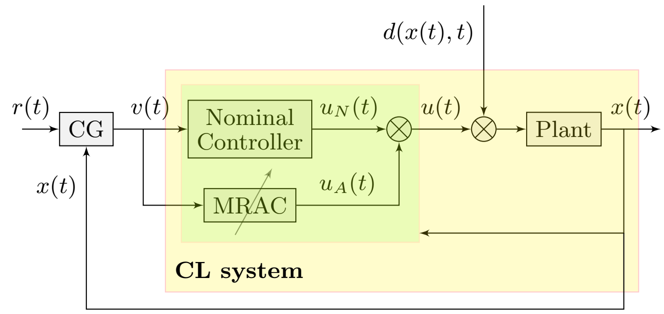

Fig. 1: CG-MRAC control strategy.

A. MRAC design

The working principle of MRAC consists in regulating the error between a desired, LTI, stable dynamics, and a non-linear, possibly unstable, time-variant, partially/completely unknown plant. When the linear-dominant dynamics of the plant are available, as in (1), the desired dynamics to be tracked can be chosen to be those achieved by a linear controller in nominal conditions, *i.e.*, assuming $d(x(t), t) = 0$. For the sake of clarity and without loss of generality, one can consider such a controller to be composed of a feedback law and a feedforward law defined by the static gains $K_x \in \mathbb{R}^{m\times n}$ and $K_r \in \mathbb{R}^{m\times r}$, respectively. Then, the nominal control law reads

$$u_N(t) = -K_x x(t) + K_r r(t). \tag{5}$$

Accordingly, the desired reference dynamics read

$$\begin{aligned} \dot{x}_r(t) &= Ax_r(t) + Bu_N(t) \\ &= A_r x_r(t) + B_r r(t), \end{aligned} \tag{6}$$

where $x_r(t) \in \mathbb{R}^n$ describes the reference state that we desire to track. The matrices $A_r \triangleq (A - BK_x) \in \mathbb{R}^{n\times n}$ and $B_r \triangleq BK_r \in \mathbb{R}^{n\times r}$ define the desired dynamics that the plant would achieve in nominal conditions. Specifically, to ensure the system stability, the nominal control gain K_x is chosen such that the matrix A_r is Hurwitz. Finally, employing (1), (3), (4), and (5), the closed-loop system state space representation without CG, *CL system* area in Fig. 1, can be rewritten as

$$\begin{aligned} \dot{x}(t) &= Ax(t) + B(u_N(t) + u_A(t) + d(x(t), t)) \\ &= A_r x(t) + B_r r(t) + B(u_A(t) + W^\top(t)\sigma(x(t))). \end{aligned} \tag{7}$$

(7) clearly infers that the adaptive controller's role is to compensate for the unmodelled dynamics. For this scope, we look for an adaptive mechanism of the form

$$u_A(t) = -\hat{W}^\top(t)\sigma(x(t)), \tag{8}$$

where $\hat{W}(t) \in \mathbb{R}^{s\times m}$ defines an estimate of $W(t)$. If $W(t)$ is precisely estimated, then the error $e(t) = x(t) - x_r(t)$ converges to zero, and the system (1) tracks the desired dynamics defined in (6). The adaptive mechanism of the MRAC strategy is selected following the Lyapunov stability theory. A possible Lyapunov function candidate that helps retrieve an adequate signal $u_A(t)$ is given by

$$\mathcal{V}(e(t), \tilde{W}(t)) = e^\top(t)Pe(t) + \gamma^{-1}\text{tr}(\tilde{W}^\top(t)\tilde{W}(t)), \quad (9)$$

where $\tilde{W}(t) \triangleq \hat{W}(t) - W(t) \in \mathbb{R}^{s\times m}$ is the distance between the matrix W and its estimate, $\gamma \in \mathbb{R}^+$ is a user-defined parameter called *learning rate* defining the force of the adaptative mechanism, and the matrix $P \in \mathbb{R}^{n\times n}$, whose existence is ensured since A_r is Hurwitz, is a positive definite matrix solution of the Lyapunov equation

$$A_r^\top P + PA_r + Q = 0, \quad (10)$$

where $Q \in \mathbb{R}^{n\times n}$ is a positive definite matrix. Note that $\mathcal{V}(e(t), \tilde{W}(t)) > 0$ for all $(e(t), \tilde{W}(t))$ except for $(e(t), \tilde{W}(t)) = (0, 0)$, where $\mathcal{V}(0, 0) = 0$. Finally, using (6) and (7), the error dynamics are given by

$$\dot{e}(t) = A_r e(t) + B(u_A(t) + W^\top(t)\sigma(x(t))). \quad (11)$$

Hence, as in [13], the time derivative of $\mathcal{V}(e(t), \tilde{W}(t))$ is

$$\begin{aligned}
&\dot{\mathcal{V}}(e(t), \tilde{W}(t)) \\
&\quad = 2e^\top(t)P\dot{e}(t) + 2\gamma^{-1}\text{tr}(\tilde{W}^\top(t)\dot{\tilde{W}}(t)) \\
&\quad = -e^\top(t)Qe(t) - 2\gamma^{-1}\text{tr}(\tilde{W}^\top(t)\dot{W}(t)) \\
&\qquad + 2\gamma^{-1}\text{tr}(\tilde{W}^\top(t)(\gamma\sigma(x(t))e^\top(t)PB - \dot{\hat{W}}(t)).
\end{aligned} \quad (12)$$

By selecting the adaptive mechanism as

$$\dot{\hat{W}}(t) = \gamma\sigma(x(t))e^\top(t)PB, \quad t \geq 0, \quad (13)$$

we retrieve

$$\begin{aligned}
\dot{\mathcal{V}}(e(t), \tilde{W}(t)) &= -e^\top(t)Qe(t) - 2\gamma^{-1}\text{tr}(\tilde{W}^\top(t)\dot{W}(t)) \\
&\leq -\lambda_{min}(Q)||e(t)||_2^2 + 2\gamma^{-1}\tilde{w}\dot{w},
\end{aligned} \quad (14)$$

where $\tilde{w} = \hat{W}_{\max} + w$, and $\hat{W}_{\max}$ being the projection norm bound enforced by employing a $proj$ operator. Hence, $\dot{\mathcal{V}}(e(t), \tilde{W}(t)) < 0$, outside of the compact set

$$\Omega \triangleq \{(e(t), \tilde{W}(t)) : ||e(t)||_2 \leq \eta \quad \text{and} \quad ||\tilde{W}(t)||_2 \leq \tilde{w}\}, \quad (15)$$

where $\eta \triangleq \sqrt{\frac{2\gamma^{-1}w\dot{w}}{\lambda_{min}(Q)}}$. This guarantees uniform boundness of the solution $(e(t), \tilde{W}(t))$. To understand how the design parameters define the ultimate bound of the error $e(t)$ for $t \geq T$, we observe from (9) that $\lambda_{min}(P)||e(t)||_2^2 \leq \mathcal{V}(e(t), \tilde{W}(t)) \leq \lambda_{max}(P)\eta^2 + \gamma^{-1}\tilde{w}^2$, which implies that

$$||e(t)||_2 \leq \sqrt{\frac{\lambda_{max}(P)\eta^2 + \gamma^{-1}\tilde{w}^2}{\lambda_{min}(P)}}, \quad t \geq T. \quad (16)$$

This result can be easily extended to the interval for $t \in [0, T)$:

$$||e(t)||_2 \leq \sqrt{\frac{\lambda_{max}(P)||e(0)||_2 + \gamma^{-1}||\tilde{W}(0)||_2^2}{\lambda_{min}(P)}}, \quad t \in [0, T). \quad (17)$$

The error bound is inversely proportional to the learning rate γ. However, it depends as well on the bounds w and $\dot{w}$ of the unknown matrix W, resulting to be conservative and not definable at the design stage. A possible approach to ensure strict performance guarantees consists in introducing an error-dependent learning rate $\gamma(e(t))$, as in [13].
Finally, combining (8) and (13), the adaptive control law reads:

$$u_A(t) = -\gamma \int_0^t \sigma(x(\tau))e^\top(\tau)PB \, \mathrm{d}\tau \; \sigma(x(t)). \quad (18)$$

B. CG design

A governor is an add-on control block that acts as a pre-filter between an input reference signal and a closed-loop system that is proven to ensure the required performance in absence of constraints. The governor-based control scheme is illustrated in Fig. 1. In particular, the proposed strategy employs a command governor block. Assuming the discrete closed-loop system equations are available, the governor device uses the current reference $r(t)$ and the current state $x(t)$ to predict at each time step the system evolution. Then, based on the predicted trajectory, it selects the best approximation $v(t)$ of the desired reference $r(t)$ such that, if maintained constant, constraints are enforced. We introduce the maximal constraint admissible set O_∞, defined as the set gathering all couples $(v, x(t))$ for which the predicted vectors $x(k)$ and $u(k)$ respect constraints:

$$O_\infty = \{(v, x(t)) : x(k) \in \mathcal{X}, u(k) \in \mathcal{U}, \; \forall\, k \in \mathbb{Z}_+\}. \quad (19)$$

Define $\tilde{O}_\infty$ as a tightened version of O_∞ so that constraints are respected at steady-state with a nonzero margin $\epsilon > 0$, *i.e.*, $\tilde{O}_\infty = O_\infty \cap O^\epsilon$, where $O^\epsilon \triangleq \{(v, x(t)) : \overline{x}(k) \in (1-\epsilon)\mathcal{X}, \overline{u}(k) \in (1-\epsilon)\mathcal{U}\}$. Then, the governor computes the signal $v(k)$ such that $(v(k), x(k)) \in \tilde{O}_\infty$. Under the assumption that the discrete time domain matrix $\hat{A}_r$ is Shur, the constrained variables are observable, the sets $\mathcal{X}$ and $\mathcal{U}$ are compact with 0 in the interior, and ϵ is sufficiently small, then the set $\tilde{O}_\infty$ is finitely determined, see [14], which implies that there exist a finite index $N \in \mathbb{Z}_+$ such that:

$$\begin{aligned}
&\tilde{O}_\infty = \tilde{O}_N = \\
&\quad \{(v, x(t)) : x(k) \in \mathcal{X}, u(k) \in \mathcal{U}, \; \forall\, k \in [1, N]\} \cap O^\epsilon.
\end{aligned} \quad (20)$$

The working principle of the governor is shown in Fig. 2. At each time step, the role of the CG block is to select from

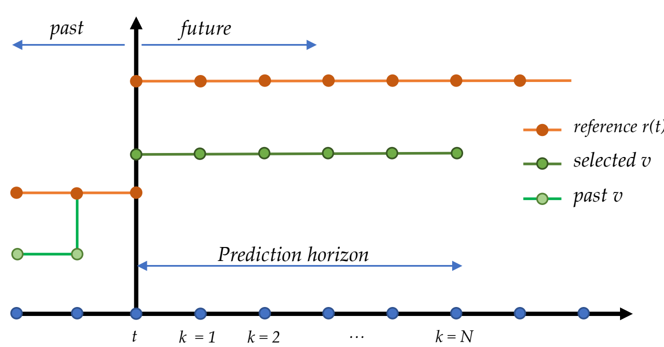

Fig. 2: Command Governor working principle.

$\tilde{O}_\infty$ the couple for which the distance between the virtual reference $v(t)$ and the desired reference $r(t)$ is minimized. For this purpose, the following optimization problem is solved:

$$v(t) = \arg\min_{\bar{v}} \|\bar{v} - r(t)\|^2 \quad \text{s.t.}(\bar{v}, x(t)) \in \tilde{O}_\infty. \tag{21}$$

The optimization problem (21) relies on the available prediction model and its accuracy determines the CG effectiveness; hence, in case the system dynamics are partially uncertain as in (1), the CG output might not ensure constraints fulfillment unless a specific treatment of the uncertainties is considered. To address this problem, several robust governors based on the available knowledge of the uncertainty magnitude bounds exist. However, their conservatism might prevent the controller from exploiting its full capacity. As in any classic robust governor, the proposed solution computes the trajectories of the nominal states $x_r(k)$ and $u_N(k)$ along the prediction horizon and ensures they remain within some tightened versions of the constraints sets $\mathcal{X}$ and $\mathcal{U}$, $\overline{\mathcal{X}}$ and $\overline{\mathcal{U}}$, respectively, see [9]. However, despite the usual solutions, the size of these sets is computed in accordance with the MRAC performance guarantees, allowing for large uncertainty handling while limiting conservatism.

Let the following set of operations be introduced:

Definition 3.1 - Let $\mathcal{P}, \mathcal{Q} \subset \mathbb{R}^n$, the Pontryagin Difference is defined as:

$$\mathcal{P} \ominus \mathcal{Q} = \{p : p + q \in \mathcal{P}, \forall q \in \mathcal{Q}\}. \tag{22}$$

Definition 3.2 - Let $\mathcal{P}, \mathcal{Q} \subset \mathbb{R}^n$, the Minkowski Sum is defined as:

$$\mathcal{P} \oplus \mathcal{Q} = \{p + q : p \in \mathcal{P}, q \in \mathcal{Q}\}. \tag{23}$$

Finally, we define as *reachable set* any set gathering all the possible evolutions of a variable.

Employing (1), (6) and the current feedback $x(t)$, the CG block sets $x(k=0) = x_r(k=0) = x(t)$, which implies $e(k=0) = 0$, and computes the reachable sets $\mathcal{F}^x_{k+1}$ of the state $x(k)$ along the horizon as

$$\mathcal{F}^x_{k+1} = \{x_r(k+1)\} \oplus \mathcal{E}, \tag{24}$$

where the uncertainty reachable set $\mathcal{E}$ takes into account the mismatch between $x(k)$ and $x_r(k)$ due to the uncertainty $d(x(t), t)$ and is built using the MRAC bounds on the error $||e(k)||_2$ defined in (17), such that $|e_i(k)| \leq ||e(k)||_2 = \sqrt{\frac{\gamma^{-1} w^2}{\lambda_{min}(P)}}, \quad i = 1, ..., n, \quad k = 1, ..., N$. Therefore, $e(k) \in \mathcal{E}$ $\forall k \in [1, N]$. Note that the size of this set does not change along the horizon, unlike in the standard CG strategy, and depends on the knowledge of the uncertainty, as well as on the learning rate γ, which is a tuning parameter. Note also that the trajectory of the nominal state $x_r(k)$ is perfectly known. Similarly, we compute the set containing all possible evolution of the control input $u(t)$ along the horizon as the sum between the reachable set $\mathcal{F}^{u_N}_{k+1}$ of the nominal control input $u_N(k)$ and the adaptive law bounds:

$$\mathcal{F}^u_{k+1} = \mathcal{F}^{u_N}_{k+1} \oplus \hat{W}_{\max}\sigma_{\max}, \tag{25}$$

where $\sigma_{\max}$ is the upper bound of $\sigma(x(t))$, assumed available, and $\mathcal{F}^{u_N}_{k+1} = -K_x \mathcal{F}^x_{k+1} + K_r r$. Then, the solution of the CG optimization problem has to ensure that $\mathcal{F}^x_{k+1} \subseteq (1-\epsilon)\mathcal{X}$ and $\mathcal{F}^u_{k+1} \subseteq (1-\epsilon)\mathcal{U}, \forall k \in [1, N]$, or, equivalently,

$$\begin{aligned} &\{x_r(k+1)\} \in \overline{\mathcal{X}} = (1-\epsilon)\mathcal{X} \ominus \mathcal{E}, \quad \text{and} \\ &\{u_N(k+1)\} \in \overline{\mathcal{U}} = (1-\epsilon)\mathcal{U} \ominus \hat{W}_{\max}\sigma_{\max}, \\ &\forall k \in [1, N]. \end{aligned} \tag{26}$$

Hence, the proposed strategy handles the mismatch between the real system evolution and the available prediction model with two main steps: tightening the constraints sets $\mathcal{X}$ and $\mathcal{U}$ to get $\overline{\mathcal{X}}$ and $\overline{\mathcal{U}}$ in accordance to the MRAC performance guarantees, and computing the known nominal trajectories of $x_r(k)$ and $u_N(t)$ along the prediction horizon to finally enforce constraints. Note that, while the second step is repeated every time step by the CG block, the computations of $\overline{\mathcal{X}}$ and $\overline{\mathcal{U}}$ are made only once at $t = 0$. Unlike the standard robust governor techniques, thanks to the MRAC law, the uncertainty reachable set $\mathcal{E}$ does not grow along the horizon. Consequently, the sets $\overline{\mathcal{X}}$ and $\overline{\mathcal{U}}$ do not shrink along the prediction, allowing handling larger uncertainties. Following these considerations, the set $\tilde{O}_\infty$ can be rewritten as:

$$\tilde{O}_\infty = \{(v, x(t)) : \, x_r(k) \in \overline{\mathcal{X}}, u_N(k) \in \overline{\mathcal{U}}, \; \forall k \in [1, N]\} \tag{27}$$

Proposition 3.1 - Let an admissible command $v(0)$ such that $(v(0), x(0)) \in \tilde{O}_\infty$ exist at $t = 0$, and $v(t)$ be constant over the prediction horizon. Then, $\tilde{O}_\infty$ is a non-empty, finitely-determined, and positively-invariant polytope.

The positive invariance of $\tilde{O}_\infty$ allows to conclude on the *recursive feasibility* of problem (21) at each time step, see [4], [9]. In addition, for a constant $r(t) = \bar{r}$, finite-time convergence to $\bar{v}$, which minimizes the norm $\|\bar{v} - r(t)\|^2$, can be proven (see [4]).

IV. NUMERICAL SIMULATION

Prime examples of constrained uncertain dynamical systems are found in space environments, which are typically subject to unmodelled phenomena that can compromise the system's reliability. Among others, satellite and rocket fuel sloshing dynamics, notably complex to model in micro-gravity, is an undeniable risk of performance degradation and in rare cases of mission loss, [15]. Model Reference Adaptive Control approaches have been shown capable of ensuring precise satellite attitude tracking in presence of multiple unmodelled dynamics; however, actuator saturations can severely jeopardize performances in case of large pointing angle maneuvers, [16]. The proposed methodology is illustrated on the plant (28) representing the simplified model of a single-axis geostationary satellite in presence of bounded exogenous disturbances:

$$\dot{x}(t) = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} x(t) + \begin{bmatrix} 0 \\ 1/J \end{bmatrix} (u(t) + d(x(t), t)), \tag{28}$$

where $x(t) = [\theta(t), \dot{\theta}(t)]^\top$ gathers the attitude and angular rate of the satellite in rad and $\mathrm{rad\,s^{-1}}$, respectively, with $x(0) = [0, 0]^\top$. Next, $J = 1000\mathrm{kg \cdot m^2}$ is the satellite

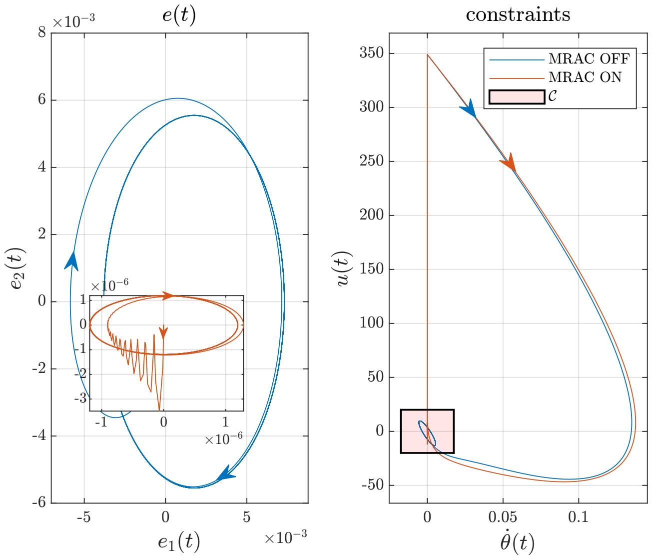

Fig. 3: Error phase portrait & constrained variables' trajectories, without VS with MRAC.

inertia matrix, and $u(t)[\mathrm{N}\cdot\mathrm{m}]$ the control input torque. Finally, the uncertainty $d(x(t),t)[\mathrm{N}\cdot\mathrm{m}]$ is parameterized in accordance to *Assumption 1* with $W(t) = [5, -2, -10sin(t)]$ and $\sigma(x(t)) = [\theta(t), \dot{\theta}(t), 1]^\top$. The system is subject to state and control input constraints:

$$|\dot{\theta}(t)| \leq \dot{\theta}_{\max}, \quad |u(t)| \leq u_{\max}, \tag{29}$$

with $\dot{\theta}_{\max} = 1\mathrm{deg}\cdot\mathrm{s}^{-1} = 0.0175\mathrm{rad}\cdot\mathrm{s}^{-1}$, and $u_{\max} = 20\mathrm{N}\cdot\mathrm{m}$. The objective is twofold: track a constant reference attitude $r = 20\mathrm{deg} = 0.3491\mathrm{rad}\ \forall t \geq 0$, while limiting the effects of the unmodelled dynamics and enforcing the system constraints.

A. MRAC Implementation

The MRAC-based controller is designed in absence of constraints. First, the nominal control law $u_N(t)$ is synthesized for the satellite rigid body dynamics, *i.e.*, $d(x(t),t) = 0$ in (28), so to impose a stable second-order closed-loop dynamics defined by $\omega_n = 1$ and $\xi = 0.9$, where ω_n and ξ are classically defined as the natural frequency and damping ratio, respectively. In accordance to (5), this yields to $K_x = [1000, 1800]$ and $K_r = 1000$. As discussed in Section III, the obtained dynamics define the matrices A_r and B_r that describe the reference model state trajectories that we wish to track by adding the adaptive control law $u_A(t)$. The MRAC parameters are set to be $\gamma = 10^{10}$ (to guarantee a strong adaptation), and $Q = I_2$. The adaptation law is then included as in (18). Fig. 3 compares the system error phase portrait with and without the adaptive law. The estimation of the weight matrix $W(t)$ allows reducing the error between the state $x(t)$ and the reference dynamics state $x_r(t)$ by three orders of magnitude. The figure also shows the evolution of the constrained states with respect to the constraints' bounds. As shown, both controllers exceed them during transient.

B. Complete Scheme Implementation

The reference dynamics equations in (6) are discretized with a sampling period of $dt = 0.02\mathrm{s}$. The prediction horizon is set to be $1\mathrm{s}$, then $N = 50$. The tightened constrained sets $\overline{\mathcal{X}}$ and $\overline{\mathcal{U}}$ are computed before running the simulation, in accordance with (26), and their size depends on the magnitude of the uncertainty and on the selected learning rate γ. In Fig. 4, we compare the standard robust command governor strategy with the proposed approach. The figure shows the constraint set $\mathcal{C}$ ($\overline{\mathcal{C}}$ in its tightened version) defining the constraints of the system. Then, we plot the predicted nominal trajectory, *i.e.*, $d(x(t),t) = 0$, defined by $\dot{\theta}_r(k)$ and $u_N(k)$ of the constrained variables $\dot{\theta}(t)$ and $u(t)$ at time $t = dt$ after the optimization problem is solved and the signal $v(t = dt)$ is selected. This signal allows constraints enforcement by simply propagating $x_r(k)$: $x(k) \in \mathcal{X}, u(k) \in \mathcal{U},\ t \geq 0$ is guaranteed by satisfying at each time step that $x_r(k) \in \overline{\mathcal{X}}, u_N(k) \in \overline{\mathcal{U}}, \forall k \in [1, N]$, for both the standard and the proposed robust command governor. In the standard robust governor approach, here built considering the static law $u(t) = u_N$ and assuming $|d(x(t),t)| \leq w\sigma_{\max} = ||W(t)||_2||\sigma([0.4, \dot{\theta}_{\max}]^\top)||_2$, the constrained set $\overline{\mathcal{C}}$ shrinks along the horizon to handle the uncertainty propagation. Instead, the MRAC properties allow the set $\overline{\mathcal{C}}$ to remain fixed along the prediction horizon, making the optimization problem lighter in terms of computational cost. Note that in the proposed strategy the set $\overline{\mathcal{C}}$ is strongly tightened on $u(k)$ to let the adaptive signal $u_A(t)$ compensate properly for the uncertainty. Note also that the proposed strategy permits handling larger uncertainties and employing a larger prediction horizon that would make the sets $\overline{\mathcal{C}}$ of the standard CG shrink excessively.

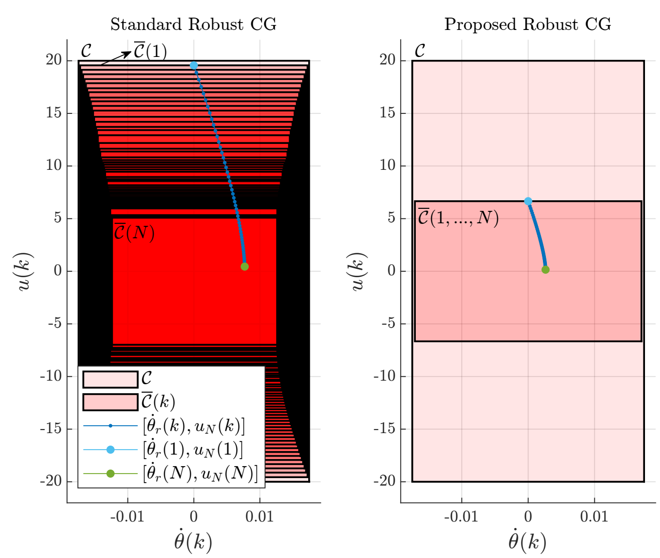

Fig. 4: Prediction of $\dot{\theta}_r(k)$ and $u_N(k)$ at time $t = dt$. Comparison between the standard Robust CG and the proposed Robust CG.

In Fig. 5 and Fig. 6, we compare the results of the two governors on a 40s simulation. Constraints are enforced by simply modifying the desired reference $r(t)$ into a virtual reference signal $v(t)$, which eventually converges to $r(t)$, see Fig. 5. The adaptive law allows converging to the reference more precisely and in a shorter time while respecting constraints. In Fig. 6, we compare the trajectories of the two constrained states. Despite the fact that the standard CG gets closer to the boundaries of the constraints, it fails in handling the uncertainty and cannot guarantee precise tracking. For what concerns the proposed strategy, the figure suggests that conservatism can be reduced, which will be the scope of future works.

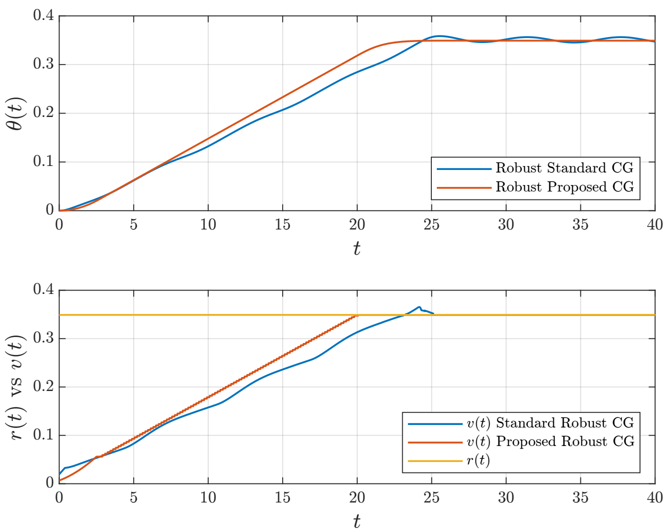

Fig. 5: $\theta(t)$ and $v(t)$. Comparison between the Standard Robust CG and the Proposed Robust CG.

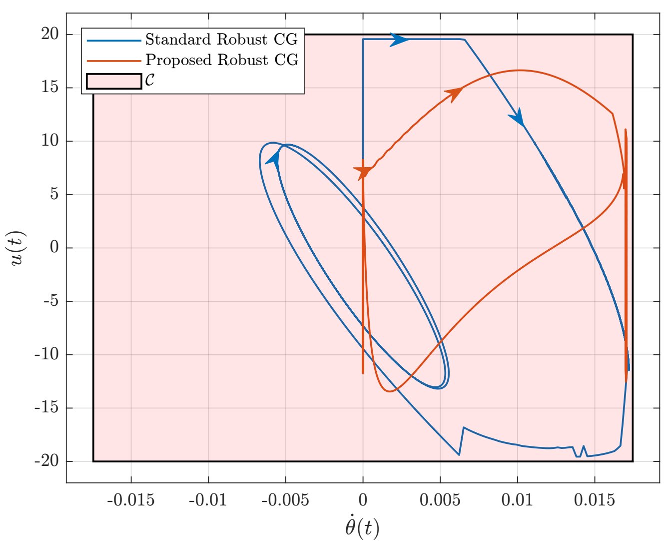

Fig. 6: Constrained variables' trajectories. Comparison between the Standard Robust CG and the Proposed Robust CG.

V. CONCLUSIONS

In this paper, a control framework for constrained linear systems with state- and time-dependent matched uncertainties is proposed. The solution combined a model reference adaptive control law with a robust command governor. The prior allowed for precise reference tracking despite the unmodelled dynamics, the latter enforced the state and input constraints by solving an optimization problem at each time step. Compared to the standard robust governor, thanks to the performance guarantees of the adaptive law, the proposed solution is shown capable of handling larger disturbances with low conservatism. The controller implementation is relatively simple and straightforward. Recursive feasibility and closed-loop system stability are established. Future works will implement advanced adaptive laws to decouple the performance guarantees from the knowledge of the uncertainty and build advanced governor schemes accordingly. Finally, experimental validation will be considered.

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